VendorPulse: 3-Minute Video Demonstration Script

ModelX 2.0 Video Presentation & Live System Walkthrough

Presenter: Sahas Abeygunarathne | **Target Duration:** Exactly 180 seconds (3:00 min)

0:00 - 0:40 | Part 1: The Predatory Debt Spiral (Problem Framing)

Visual: Footage of Colombo Pettah street vendors and fruit hawkers along Bodhiraja Mawatha. Monsoon clouds gathering overhead.

"Narration: Across urban commercial hubs like Colombo, tens of thousands of informal street vendors and daily-wage hawkers survive on day-to-day liquidity. Today's sales purchase tomorrow's produce. When localized shocks strike—such as monsoon flash flooding or unannounced transit strikes—pedestrian foot traffic halts instantly. Without cash buffers, vendors face an impossible choice: close down, or borrow emergency working capital from local predatory lenders charging ruinous 10% to 20% weekly interest. Traditional micro-finance programs react weeks too late. Meet VendorPulse: an early-warning platform shifting NGO coordinators from reactive recovery to proactive financial shielding."

0:40 - 1:20 | Part 2: Multi-Modal Ingestion & Intelligence Core

Visual: Split-screen showing live OpenWeatherMap forecast, OpenStreetMap corridor telemetry (64% speed drop), and spaCy NLP extraction of 'Bus Route 138 suspended'.

"Narration: VendorPulse operates by algorithmically extracting compounded risk from three disparate urban data streams. First, it pulls hyper-local 72-hour precipitation forecasts from OpenWeatherMap. Second, it monitors corridor speed drops using OpenStreetMap routing. Third, our localized NLP engine scans civic news feeds, isolating Colombo landmarks like Bodhiraja Mawatha and technical bus suspensions. A multi-horizon neural model trained on two years of historical Colombo disruption data then calculates the probability of cash-flow collapse. While moderate rain or minor traffic delays alone are manageable, our model detects when their compounded interaction triggers a catastrophic cliff-edge."

1:20 - 2:20 | Part 3: Live Coordinator Experience & 1-Click Bridge Deployment

Visual: Screen recording of the dark-mode Threat Map in action. Coordinator selects Pettah Intersection (Block P-04), reviews 90% confidence, 142 vendors, and Rs 2.5M capital, and taps 'Authorize Bridge Deployment'.

"Narration: Here is how an NGO Field Outreach Coordinator interacts with VendorPulse on a Thursday morning. Opening the live Threat Map, they immediately spot a critical red alert over Pettah Block P-04. Tapping into the zone card reveals the full predictive breakdown: 48 hours before Friday trading, the system identifies a 90% probability of severe cash-flow failure. The coordinator sees that 142 registered vendors are at risk, requiring Rs 2.5 million in bridge capital. Below, the live drivers confirm the compounded cause: primary drainage overflowing by 52 centimeters coupled with the total suspension of Bus Route 138. With one tap on 'Authorize Bridge Deployment', the coordinator dispatches zero-interest emergency bridge loans, logging the audit record directly to Supabase and notifying ground field officers before predatory lenders can intervene."

2:20 - 3:00 | Part 4: Empirical Evaluation, Baselines & Social Impact

Visual: On-screen evaluation table showing F1: 0.773 vs Naive 0.122 vs ARIMA 0.000, and FPR 2.3% < 20%. Closing credits with Team Nexio and IEEE CIS IIT.

"Narration: We rigorously evaluated VendorPulse against the project proposal's criteria on a 600-day out-of-sample holdout dataset. VendorPulse achieved an F1-score of 0.773, drastically outperforming the Naïve persistence baseline of 0.122 and the traditional ARIMA model, which scored 0.000 because linear models cannot anticipate sudden non-linear shocks. Crucially, our False Positive Rate was just 2.3%—far below the 20% failure threshold—ensuring NGO capital is never squandered on false alarms. By transforming lagging relief into proactive shielding, VendorPulse protects urban informal vendors from debt traps, safeguarding livelihoods one street block at a time. Thank you."